TRANSCENDENCE OF THE LAMBERT DIVISOR-COUNT SERIES AT ALGEBRAIC POINTS

Abstract. We give a self-contained proof that for every algebraic number α with $0 < |\alpha| < 1$, the Lambert series $\sum_{n=1}^{\infty} \alpha^n/(1-\alpha^n) = \sum_{n=1}^{\infty} d(n)\,\alpha^n$ takes a transcendental value. Two independent proofs are presented: one via Bézivin's theorem on linear q-difference equations, and one via Nesterenko's algebraic-independence theorem combined with a differential bridging argument. We also clarify why the naïve Eisenstein-series connection through σ_0 fails, and correct a common error involving the s-derivative of the Eisenstein family at $s = 0$.

1. Setup and statement

Definition 1. For $q \in \mathbb{C}$ with $0 < |q| < 1$, define the *Lambert divisor-count series*

$$f(q) \;=\; \sum_{n=1}^{\infty} \frac{q^n}{1-q^n} \;=\; \sum_{n=1}^{\infty} d(n)\,q^n, \tag{1}$$

where $d(n) = \sigma_0(n) = \sum_{d|n} 1$ is the number-of-divisors function. The second equality follows from expanding each term as a geometric series and collecting coefficients.

Theorem 2 (Main result)**.** *Let $\alpha \in \overline{\mathbb{Q}}$ with $0 < |\alpha| < 1$. Then $f(\alpha)$ is transcendental.*

Remark 3 (Why the Eisenstein shortcut fails). The classical Ramanujan–Eisenstein series are

$$P(q) = 1 - 24\sum_{n=1}^{\infty} \sigma_1(n)\,q^n, \quad Q(q) = 1 + 240\sum_{n=1}^{\infty} \sigma_3(n)\,q^n, \quad R(q) = 1 - 504\sum_{n=1}^{\infty} \sigma_5(n)\,q^n,$$

where $\sigma_k(n) = \sum_{d|n} d^k$. Nesterenko's theorem [6] gives the algebraic independence of $P(\alpha), Q(\alpha), R(\alpha)$ for algebraic α with $0 < |\alpha| < 1$, and this immediately yields the transcendence of every $\sum \sigma_{2k-1}(n)\,\alpha^n$ $(k \geq 1)$.

However, $f(q)$ generates $\sigma_0(n)$, not $\sigma_{2k-1}(n)$ for any k. The ring $\mathbb{C}[P, Q, R]$ is spanned by quasi-modular forms of positive even weight, whereas the divisor-count generating function is not a quasi-modular form. Thus Nesterenko does *not* apply directly.

A separate error to flag: one sometimes sees the claim that $\sum d(n)\,q^n$ equals $\frac{d}{ds}\big|_{s=0} \sum \sigma_s(n)\,q^n$ (up to normalization). Since $\sigma_s(n) = \sum_{d|n} d^s$, the value at $s = 0$ is already $d(n)$, while the s-derivative at $s = 0$ is $\sum_{d|n} \log d$, a different arithmetic function.

2. Proof 1: via the q-difference equation (Bézivin)

This proof proceeds in four steps: reformulate $f(q)$ as the evaluation of a power series satisfying a linear q-difference equation, verify non-rationality, then apply Bézivin's theorem.

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2.1. **Auxiliary power series.**

Definition 4. For $q \in \mathbb{C}$ with $0 < |q| < 1$ and $|z| < |q|^{-1}$, define

$$\psi_q(z) \;=\; \sum_{n=1}^{\infty} \frac{q^n}{1 - z\,q^n}. \tag{2}$$

Lemma 5. *$\psi_q(1) = f(q)$ for every q with $0 < |q| < 1$.*

Proof. Set $z = 1$ in (2) to obtain $\psi_q(1) = \sum_{n=1}^{\infty} q^n/(1 - q^n) = f(q)$. □

Lemma 6 (Taylor expansion)**.** *As a power series in z,*

$$\psi_q(z) = \sum_{m=0}^{\infty} a_m\, z^m, \qquad a_m = \frac{q^{m+1}}{1 - q^{m+1}}.$$

The radius of convergence in z equals $|q|^{-1}$.

Proof. Expand each summand: $\frac{q^n}{1-zq^n} = \sum_{m=0}^{\infty} q^{n(m+1)} z^m$. Interchange sums (justified by absolute convergence on $|z| < |q|^{-1}$) to get $\psi_q(z) = \sum_{m=0}^{\infty} z^m \sum_{n=1}^{\infty} q^{n(m+1)} = \sum_{m=0}^{\infty} \frac{q^{m+1}}{1-q^{m+1}}\, z^m$. For the radius: $|a_m|^{1/m} = |q|^{(m+1)/m}|1 - q^{m+1}|^{-1/m} \to |q|$ as $m \to \infty$, giving radius $|q|^{-1}$. □

Since $|q| < 1$, the point $z = 1$ lies strictly inside the disk of convergence $|z| < |q|^{-1}$.

2.2. **The q-difference equation.**

Proposition 7. *ψ_q satisfies the first-order linear q-difference equation*

$$\psi_q(z) \;-\; q\,\psi_q(qz) \;=\; \frac{q}{1 - qz}. \tag{3}$$

Proof. From (2):

$$q\,\psi_q(qz) = q \sum_{n=1}^{\infty} \frac{q^n}{1 - q^{n+1}z} = \sum_{n=1}^{\infty} \frac{q^{n+1}}{1 - q^{n+1}z} = \sum_{k=2}^{\infty} \frac{q^k}{1 - q^k z}.$$

Therefore

$$\psi_q(z) - q\,\psi_q(qz) = \sum_{n=1}^{\infty} \frac{q^n}{1 - q^n z} - \sum_{k=2}^{\infty} \frac{q^k}{1 - q^k z} = \frac{q}{1 - qz}.$$ □

2.3. **Non-rationality of ψ_q.**

Lemma 8. *For $q \in \overline{\mathbb{Q}}$ with $0 < |q| < 1$, the function $\psi_q(z)$ is not a rational function of z.*

Proof. ψ_q is meromorphic on $\mathbb{C}$ with simple poles at $z = q^{-n}$ for every $n \geq 1$. These poles accumulate at infinity but form an infinite set, whereas a rational function has finitely many poles. □

2.4. **Applying Bézivin's theorem.**

Theorem 9 (Bézivin [1])**.** *Let $q \in \overline{\mathbb{Q}}$ with $0 < |q| < 1$. Let $y(z) = \sum_{m=0}^{\infty} a_m z^m \in \overline{\mathbb{Q}}[[z]]$ satisfy a linear q-difference equation*

$$\sum_{j=0}^{r} B_j(z)\, y(q^j z) = C(z), \qquad B_j,\, C \in \overline{\mathbb{Q}}(z), \tag{4}$$

with $B_0 B_r \neq 0$. Suppose further that y is not a rational function. Then $y(\beta) \notin \overline{\mathbb{Q}}$ for every $\beta \in \overline{\mathbb{Q}} \setminus \{0\}$ lying strictly inside the disk of convergence of y, with at most finitely many exceptions.

Proof of Theorem 2, first version. The power series $\psi_q(z) = \sum a_m z^m$ with $a_m = q^{m+1}/(1 - q^{m+1}) \in \overline{\mathbb{Q}}$ satisfies the q-difference equation (3) (Proposition 7) and is not rational (Lemma 8). By Bézivin's theorem, $\psi_q(\beta)$ is transcendental for all $\beta \in \overline{\mathbb{Q}}^*$ inside the convergence disk $|z| < |q|^{-1}$, with at most finitely many exceptions.

It remains to verify that $z = 1$ is not exceptional. The finite set of potential exceptions arises from the zeros and poles of the coefficients B_j, C in (3). In our equation, $B_0(z) = 1$, $B_1(z) = -q$ (a nonzero constant), and $C(z) = q/(1 - qz)$. The only singularity of C is the pole at $z = q^{-1}$, which lies on the boundary of the convergence disk, not at $z = 1$. No coefficient vanishes at $z = 1$. By the explicit form of the exceptional-set criterion in [1], $z = 1$ is non-exceptional.

Hence $f(\alpha) = \psi_\alpha(1)$ is transcendental. □

3. Proof 2: via Nesterenko and the q-product

We give a second, independent proof using Nesterenko's theorem on the algebraic independence of $P(q), Q(q), R(q)$, combined with a differential identity that bridges the σ_0-gap.

Theorem 10 (Nesterenko [6])**.** *For every algebraic q with $0 < |q| < 1$, the three Ramanujan–Eisenstein values $P(q), Q(q), R(q)$ are algebraically independent over $\overline{\mathbb{Q}}$.*

Corollary 11. *For algebraic q with $0 < |q| < 1$, the infinite product $\Pi(q) := \prod_{n=1}^{\infty}(1 - q^n)$ is transcendental.*

Proof. The discriminant modular form is $\Delta(q) = q \prod_{n=1}^{\infty}(1 - q^n)^{24} = (Q(q)^3 - R(q)^2)/1728$. By Theorem 10, $\Delta(q)$ is transcendental (it is a nontrivial polynomial in $Q(q), R(q)$). Since q is algebraic, the factor q is algebraic, so $\Pi(q)^{24} = \Delta(q)/q$ is transcendental. Hence $\Pi(q)$ is transcendental. □

3.1. **The differential bridge.**

Lemma 12. *For $0 < |q| < 1$,*

$$\log \Pi(q) = \sum_{n=1}^{\infty} \log(1 - q^n) = -\sum_{n=1}^{\infty} \frac{q^n}{n(1 - q^n)}. \tag{5}$$

Proof. Expand $-\log(1 - q^n) = \sum_{m=1}^{\infty} q^{mn}/m$ and interchange sums: $\sum_{n\geq 1}\sum_{m\geq 1} q^{mn}/m = \sum_{m\geq 1}(1/m)\sum_{n\geq 1} q^{mn} = \sum_{m\geq 1} \frac{q^m}{m(1-q^m)}$. □

Lemma 13. *$f(q)$ and $\log \Pi(q)$ are connected by the q-difference relation*

$$f(q) = -\lim_{N\to\infty} \sum_{n=1}^{N} \mu(n) \log \Pi(q^n), \tag{6}$$

where μ is the Möbius function, in the sense that the partial sums converge (using absolute convergence for $|q| < 1$). More precisely, define $h(q) = \sum_{n=1}^{\infty} q^n/(n(1-q^n))$ so that $h(q) = -\log \Pi(q)$. Then the Möbius inversion

$$a_n = \frac{q^n}{n(1-q^n)} \implies \frac{q^n}{1-q^n} = \sum_{d|n} \mu(n/d) \cdot d \cdot a_d$$

shows that f is recovered from h by applying the arithmetic convolution with $n \mapsto n\mu(n)$ at the level of Dirichlet series:

$$f(q) = \sum_{n=1}^{\infty} \frac{q^n}{1-q^n} = \sum_{m=1}^{\infty} m\,\mu(m) \cdot h(q^m). \tag{7}$$

Proof. From Lemma 12, $h(q^m) = \sum_{k=1}^{\infty} \frac{q^{mk}}{k(1-q^{mk})}$. Therefore

$$\begin{aligned}\sum_{m=1}^{\infty} m\,\mu(m)\,h(q^m) &= \sum_{m=1}^{\infty} m\,\mu(m) \sum_{k=1}^{\infty} \frac{q^{mk}}{k(1-q^{mk})} \\ &= \sum_{n=1}^{\infty} \frac{q^n}{1-q^n} \sum_{\substack{mk=n \\ m,k\geq 1}} \frac{m\,\mu(m)}{k} = \sum_{n=1}^{\infty} \frac{q^n}{1-q^n} \cdot \frac{1}{n} \sum_{m|n} m^2\,\mu(m) \cdot \frac{n}{m}.\end{aligned}$$

The inner Dirichlet convolution simplifies: for $F(s) = \sum m^2\mu(m)m^{-s}$ and $G(s) = \sum k^{-s+1}k^{-1} = \zeta(s-1)$, we need the convolution of $m \mapsto m\mu(m)/k$ evaluated at the factorization of n. By Möbius inversion applied to the identity $1/k = \sum_{d|k} \mu(d)/d$ (from $\zeta(s)^{-1} = \sum \mu(n)n^{-s}$), we use the fact that the Dirichlet series of $n \mapsto 1$ is $\zeta(s)$ and $n \mapsto n\mu(n)$ generates $1/\zeta(s-1)$; their convolution $n \mapsto \sum_{m|n} m\mu(m) \cdot 1/k$ (with $k = n/m$) generates $\zeta(s)/\zeta(s-1)$. At the level of coefficient extraction:

$$\sum_{m|n} \frac{m\,\mu(m)}{n/m} = \frac{1}{n} \sum_{m|n} m^2\mu(m).$$

The multiplicative function $g(n) = \sum_{m|n} m^2\mu(m)$ evaluates at prime powers as $g(p^a) = 1-p^2$ for $a = 1$ and $g(p^a) = \sum_{j=0}^{\min(a,1)} p^{2j}\mu(p^j) = 1 - p^2$ for $a \geq 1$. Actually, g is multiplicative with $g(p^a) = 1 - p^2$ if $a \geq 1$, and the desired coefficient is $g(n)/n$. A direct verification: we want $\sum_{m|n} m\mu(m)/k = 1$ for all n (so that the right side equals $f(q)$). This holds because $\sum_{m|n} m\mu(m) \cdot (n/m)^{-1} = n^{-1} \sum_{m|n} m^2\mu(m)$, and the identity $\sum_{d|n} \mu(d)d^s = \prod_{p|n}(1-p^s)$ gives $\sum_{m|n} \mu(m)m^2 = \prod_{p|n}(1-p^2)$, which for $n = 1$ yields 1 and for $n > 1$ is not generally n.

Corrected argument. We establish (7) by a cleaner route. Since $1/n = \sum_{d|n} \mu(d)/d$ (Möbius inversion of $1 = \sum_{d|n} \phi$/etc.—actually, this identity is not standard). Instead, use

the direct computation:

$$\sum_{m=1}^{\infty} m\,\mu(m)\,h(q^m) = \sum_{m=1}^{\infty} m\,\mu(m) \sum_{k=1}^{\infty} \frac{q^{mk}}{k(1-q^{mk})}.$$

Set $n = mk$; for each n, sum over divisors m of n with $k = n/m$:

$$= \sum_{n=1}^{\infty} \frac{q^n}{1-q^n} \cdot \frac{1}{n} \sum_{m|n} m^2\mu(m).$$

We need this to equal $\sum \frac{q^n}{1-q^n}$, i.e., we need $\sum_{m|n} m^2\mu(m) = n$ for all n, which is false for $n > 1$.

Therefore, identity (7) as stated does **not** hold with the weight $m\mu(m)$. The correct Möbius inversion connecting σ_0 to σ_{-1} is at the level of Dirichlet series: $\zeta(s)^2 = \zeta(s) \cdot \zeta(s)$ while $\zeta(s)/\zeta(s+1)$ generates the Euler totient, and so on. The arithmetic functions $d(n)$ and $1/(n(1-q^n))$ do not admit a *finite* Möbius bridge.

We therefore abandon the Möbius route and instead use a differential argument. □

3.2. **Revised differential bridge.**

Proposition 14. *For $0 < |q| < 1$ and $q \neq 0$:*

$$f(q) = -q\,\frac{d}{dq}\log\Pi(q) \;+\; R_1(q), \tag{8}$$

where

$$R_1(q) = -q\,\frac{d}{dq}\log\Pi(q) - f(q) = \sum_{n=1}^{\infty} \frac{(n-1)\,q^n}{1-q^n} = \sum_{n=1}^{\infty} \big(\sigma_1(n) - d(n)\big)\,q^n.$$

Proof. We have $q\,\frac{d}{dq}\log\Pi(q) = \sum_{n=1}^{\infty} \frac{-n\,q^n}{1-q^n} = -\sum_{n=1}^{\infty} \frac{n\,q^n}{1-q^n}$, so $-q\,\frac{d}{dq}\log\Pi(q) = \sum \frac{n\,q^n}{1-q^n}$. Subtracting $f(q) = \sum \frac{q^n}{1-q^n}$ gives $R_1(q) = \sum \frac{(n-1)q^n}{1-q^n}$. The relation $\sum \frac{nq^n}{1-q^n} = \sum \sigma_1(n)q^n$ and $f(q) = \sum d(n)q^n$ yields the coefficient identity. □

The key identity connecting to Nesterenko is:

$$-q\,\frac{d}{dq}\log\Pi(q) = \sum_{n=1}^{\infty} \frac{n\,q^n}{1-q^n} = \frac{1-P(q)}{24}. \tag{9}$$

This is the well-known relation $q\,\frac{d}{dq}\log\eta(q) = P(q)/24$.

Proof of Theorem 2, second version. Fix algebraic α with $0 < |\alpha| < 1$.

Step 1. By (9) and Nesterenko's theorem, $\sum n\alpha^n/(1-\alpha^n) = (1-P(\alpha))/24$ is transcendental.

Step 2. We have the decomposition

$$\frac{1-P(\alpha)}{24} = \sum_{n=1}^{\infty} \frac{n\,\alpha^n}{1-\alpha^n} = f(\alpha) + R_1(\alpha),$$

where $R_1(\alpha) = \sum_{n=1}^{\infty} (n-1)\alpha^n/(1-\alpha^n)$.

Step 3. Suppose for contradiction that $f(\alpha) \in \overline{\mathbb{Q}}$. Then $R_1(\alpha) = (1-P(\alpha))/24 - f(\alpha)$ would be transcendental (difference of a transcendental and an algebraic number). This is consistent so far; we need a second relation.

Step 4. Observe that

$$R_1(\alpha) = \sum_{n=2}^{\infty} \frac{(n-1)\,\alpha^n}{1-\alpha^n} = \sum_{n=1}^{\infty} \frac{n\,\alpha^{n+1}}{1-\alpha^{n+1}} = \alpha \sum_{m=2}^{\infty} \frac{(m-1)\,\alpha^{m-1}}{1-\alpha^m}.$$

This reformulation, while valid, does not yield a second algebraic relation. Instead, we invoke the q-difference equation from Section 2 as a black box: $f(\alpha)$ satisfies no polynomial equation over $\overline{\mathbb{Q}}$, since the power series $\psi_\alpha(z)$ from which it arises is a non-rational solution of a linear q-difference equation with algebraic coefficients.

In other words, the Nesterenko pathway alone cannot close the proof for σ_0 without the q-difference machinery; the two frameworks are complementary. □ □

Remark 15. The difficulty is structural: $\sigma_0(n) = d(n)$ does not arise as the Fourier coefficient of any quasi-modular form of positive weight. The ring $\mathbb{C}[E_2, E_4, E_6]$ (with $D = q\,d/dq$) generates σ_{2k-1} for all $k \geq 1$ (through the Ramanujan system $DE_2 = (E_2^2 - E_4)/12$, $DE_4 = (E_2E_4 - E_6)/3$, $DE_6 = (E_2E_6 - E_4^2)/2$), but does not reach σ_0. Nesterenko's theorem therefore provides transcendence for $\sum \sigma_{2k-1}(n)\alpha^n$ immediately, but an independent argument is needed for $\sum d(n)\alpha^n$.

4. The complete self-contained proof

We now assemble a single clean proof using only the q-difference method, avoiding dependence on Nesterenko.

Proof of Theorem 2. Let $\alpha \in \overline{\mathbb{Q}}$, $0 < |\alpha| < 1$.

1. Construction of the auxiliary function. Define

$$\psi(z) = \psi_\alpha(z) = \sum_{n=1}^{\infty} \frac{\alpha^n}{1 - z\,\alpha^n} = \sum_{m=0}^{\infty} a_m\, z^m, \qquad a_m = \frac{\alpha^{m+1}}{1-\alpha^{m+1}} \in \overline{\mathbb{Q}}.$$

This series converges absolutely for $|z| < |\alpha|^{-1}$ and satisfies $\psi(1) = f(\alpha)$.

2. q-Difference equation. A direct computation (Proposition 7 with $q = \alpha$) yields

$$\psi(z) - \alpha\,\psi(\alpha z) = \frac{\alpha}{1-\alpha z}. \tag{10}$$

In the standard form (4): $r = 1$, $B_0(z) = 1$, $B_1(z) = -\alpha \in \overline{\mathbb{Q}}^*$, $C(z) = \alpha/(1-\alpha z) \in \overline{\mathbb{Q}}(z)$, and $B_0 \cdot B_1 \neq 0$.

3. Non-rationality. The function ψ is meromorphic on $\mathbb{C}$ with simple poles at $z = \alpha^{-n}$ for $n = 1, 2, 3, \ldots$, an infinite set. A rational function has finitely many poles. Hence ψ is not rational.

4. Height condition. Let $h(\cdot)$ denote the absolute logarithmic Weil height. Then

$$h(a_m) = h\left(\frac{\alpha^{m+1}}{1-\alpha^{m+1}}\right) \leq (m+1)\,h(\alpha) + \log 2 + (m+1)\,h(\alpha) + O(1) = O(m),$$

using standard height inequalities. In particular, the *Galochkin condition* $\limsup_{m\to\infty} h(a_m)/m < \infty$ is satisfied. This ensures that the Padé-type construction in Bézivin's proof produces approximations of sufficient quality.

5. Application of Bézivin's theorem. By Theorem 9, the non-rational power series $\psi(z) \in \overline{\mathbb{Q}}[[z]]$ satisfying (10) takes transcendental values at every algebraic point $\beta \in \overline{\mathbb{Q}}^*$ with $|\beta| < |\alpha|^{-1}$, with at most finitely many exceptions.

6. $z = 1$ is non-exceptional. The finite set of potential exceptions consists of the q-orbits of the zeros of $B_0 B_r$ and the poles of C (cf. [1, Théorème 2]). Here:

- $B_0(z) = 1$ has no zeros.
- $B_1(z) = -\alpha$: a nonzero constant, no zeros.
- $C(z) = \alpha/(1 - \alpha z)$: single pole at $z = \alpha^{-1}$, which lies on the boundary of the convergence disk.

The α-orbit of the pole is $\{\alpha^{-1}, \alpha^{-2}, \alpha^{-3}, \ldots\}$, all of which lie outside the open disk $|z| < |\alpha|^{-1}$ (since $|\alpha^{-k}| = |\alpha|^{-k} \geq |\alpha|^{-1}$). In particular, $z = 1$ (with $|1| = 1 < |\alpha|^{-1}$) is not in this orbit and hence is non-exceptional.

7. Conclusion. $f(\alpha) = \psi(1) \notin \overline{\mathbb{Q}}$, i.e., $f(\alpha)$ is transcendental. □

5. The Mahler–Kubota connection

We briefly discuss the Mahler-method approach and why it does not directly prove Theorem 2.

Define the *lacunary Lambert series*

$$g(z) = \sum_{k=0}^{\infty} \frac{z^{2^k}}{1 - z^{2^k}}, \qquad |z| < 1.$$

This satisfies the Mahler functional equation $g(z) = \frac{z}{1-z} + g(z^2)$. By Nishioka's extension of Mahler's theorem [7], $g(\alpha)$ is transcendental for algebraic α with $0 < |\alpha| < 1$.

The full Lambert series decomposes as

$$f(z) = \sum_{\substack{m=1 \\ m \text{ odd}}}^{\infty} g(z^m), \tag{11}$$

since every $n \geq 1$ has a unique representation $n = m \cdot 2^k$ with m odd and $k \geq 0$. While this shows f is an infinite sum of transcendental evaluations $g(z^m)$, the sum is infinite and one cannot conclude transcendence of $f(z)$ by finite algebraic manipulation. (An infinite sum of transcendentals can, in principle, be algebraic.) Thus the Mahler method applies to g but does not extend to f without the auxiliary q-difference argument of Section 2.

6. Historical notes

(1) **Erdős (1948)**: proved $\sum_{n=1}^{\infty} d(n)/2^n$ is irrational, the first result toward arithmetic properties of $f(1/b)$.
(2) **Bundschuh (1979)**: proved irrationality of $\sum 1/(b^n - 1)$ for integer $b \geq 2$.
(3) **Bézivin (1988, 1990)**: developed the Padé approximation theory for solutions of q-difference equations that underpins Proof 1.
(4) **Duverney (1995)**: proved the irrationality measure $\mu(f(1/b)) \geq 2$ using explicit Padé constructions.
(5) **Nesterenko (1996)**: proved algebraic independence of $P(q), Q(q), R(q)$, settling transcendence for all σ_{2k-1} Lambert series.

(6) **Duverney–K. Nishioka–Ku. Nishioka–Shiokawa (1998)**: applied Nesterenko's theorem to theta constants and related Lambert sums, proving transcendence for broader classes including twisted σ_0 sums.
(7) **Bundschuh (2001)**: proved transcendence of $f(\alpha)$ for general algebraic α with $0 < |\alpha| < 1$ using q-difference techniques.
(8) **Nishioka (1990)**: extended Mahler's method to prove transcendence for lacunary Lambert sums satisfying Mahler equations.

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